

Configuration Automation using Ansible

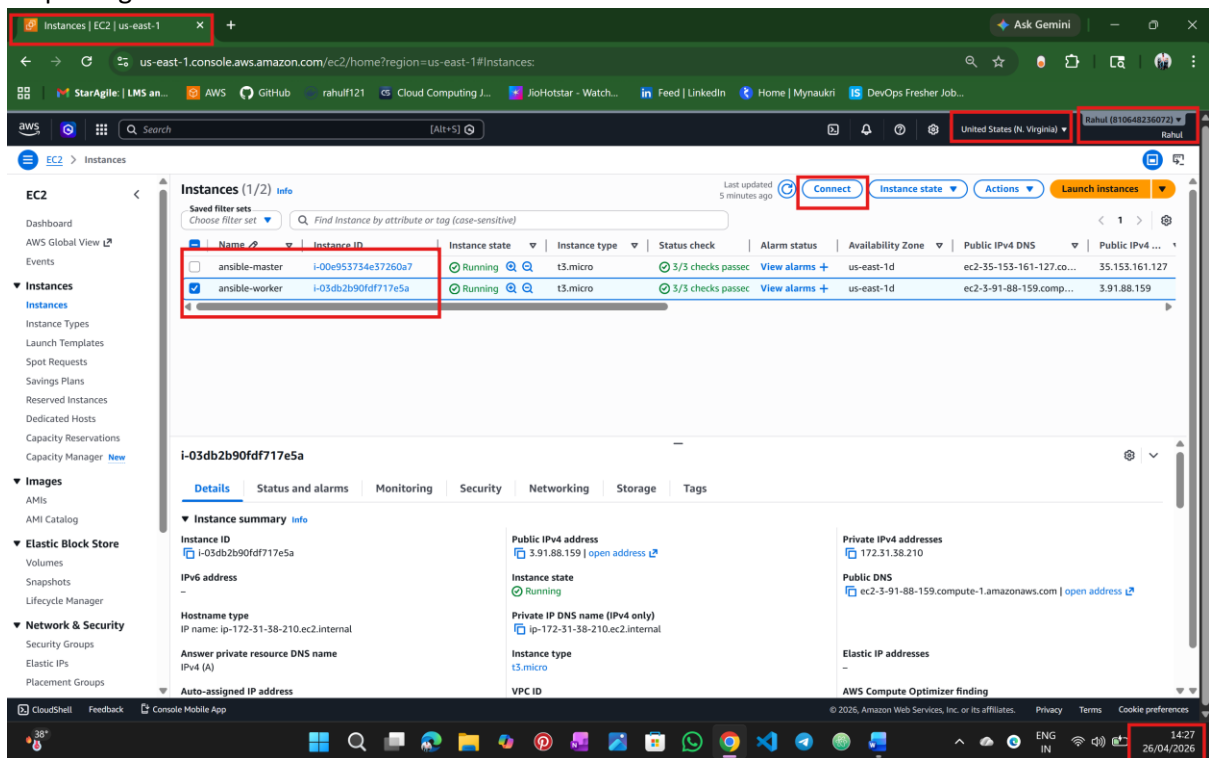
Submitted to - Vikul Sir

Date of Submission – 28/APR/2026

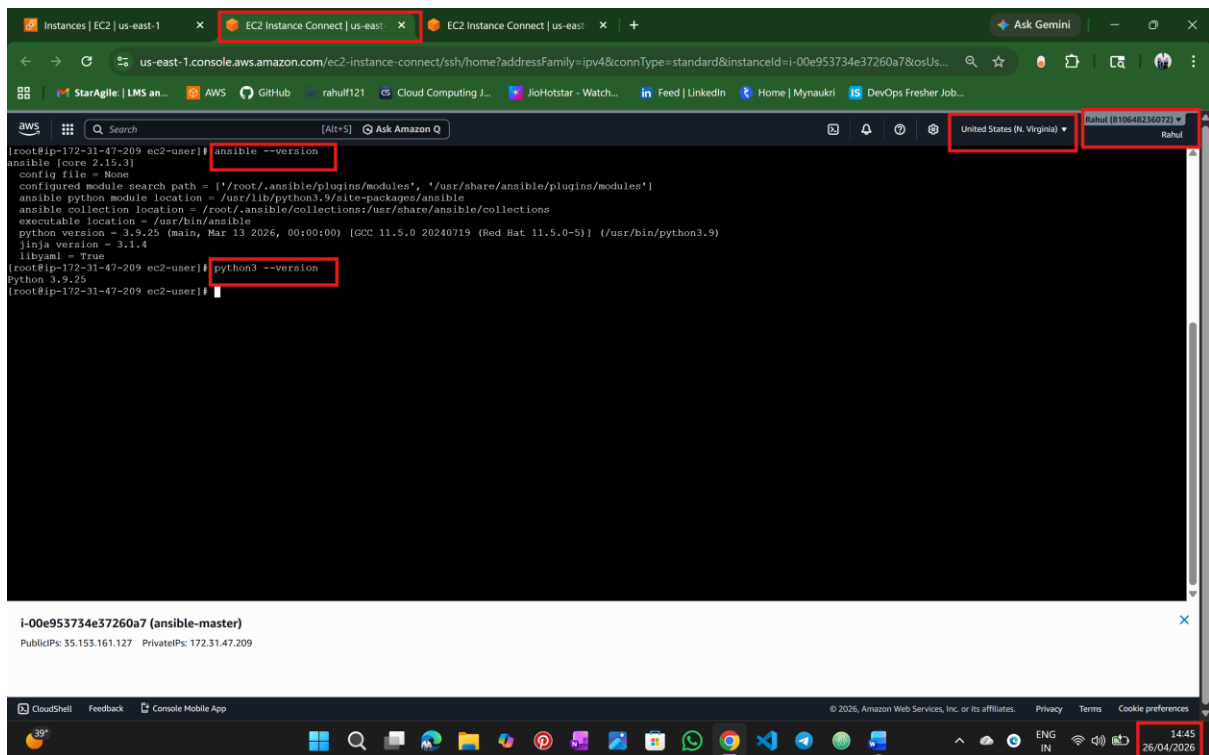
Submitted by – Rahul Nalathathi

L1 - Create and Execute Ansible Playbook to Setup Java Maven Application Build Server

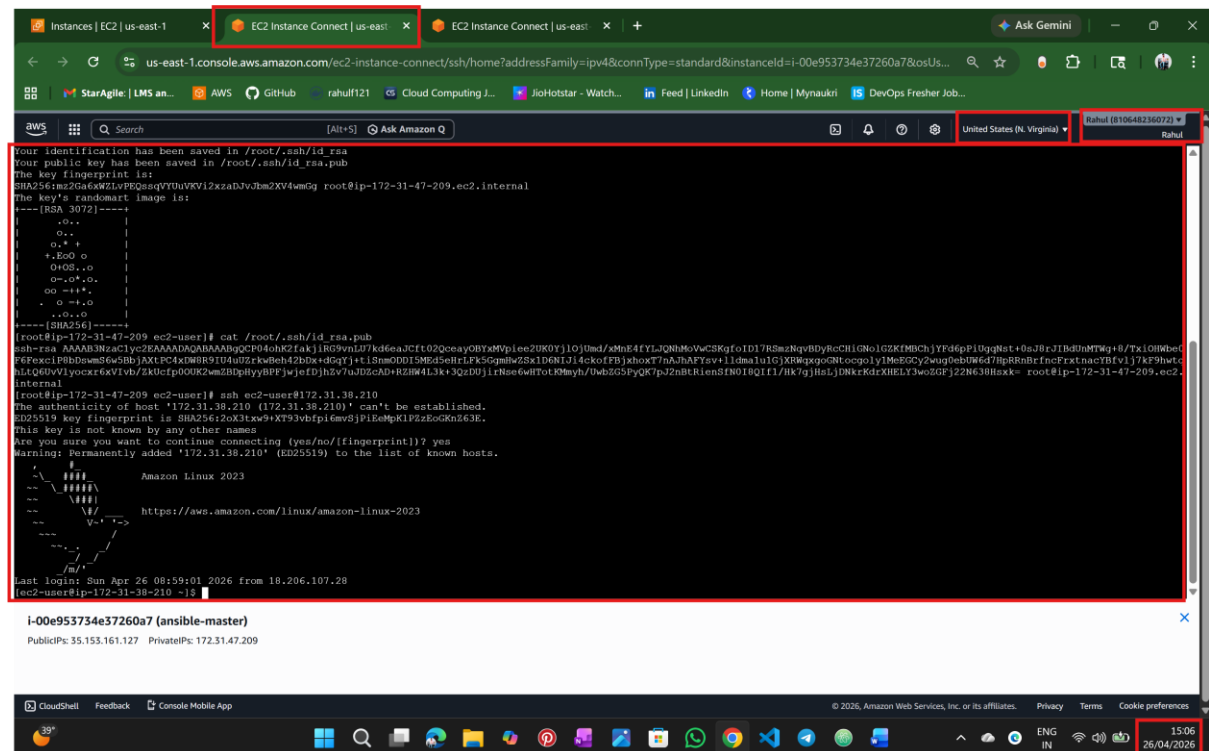
Step: 1 login into AWS console and create two instances



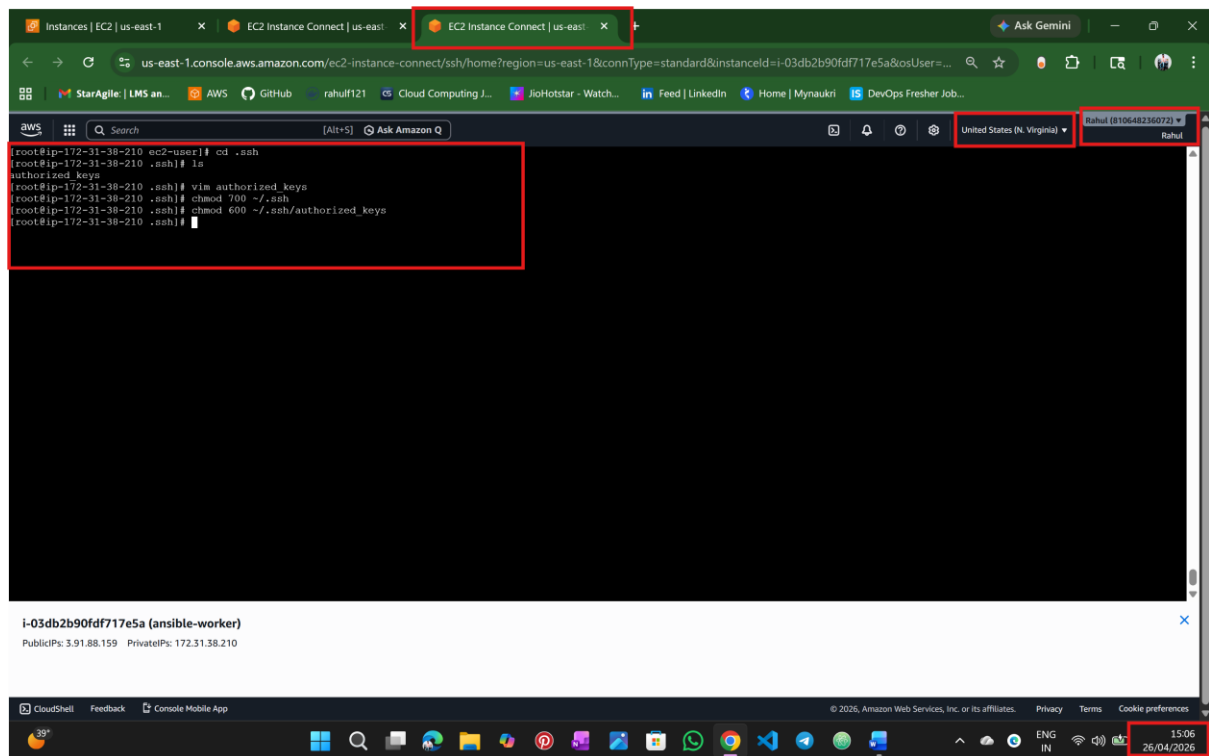
Step: 2 Master node Terminal



Step:3 connect the master node and worker node with ssh-key



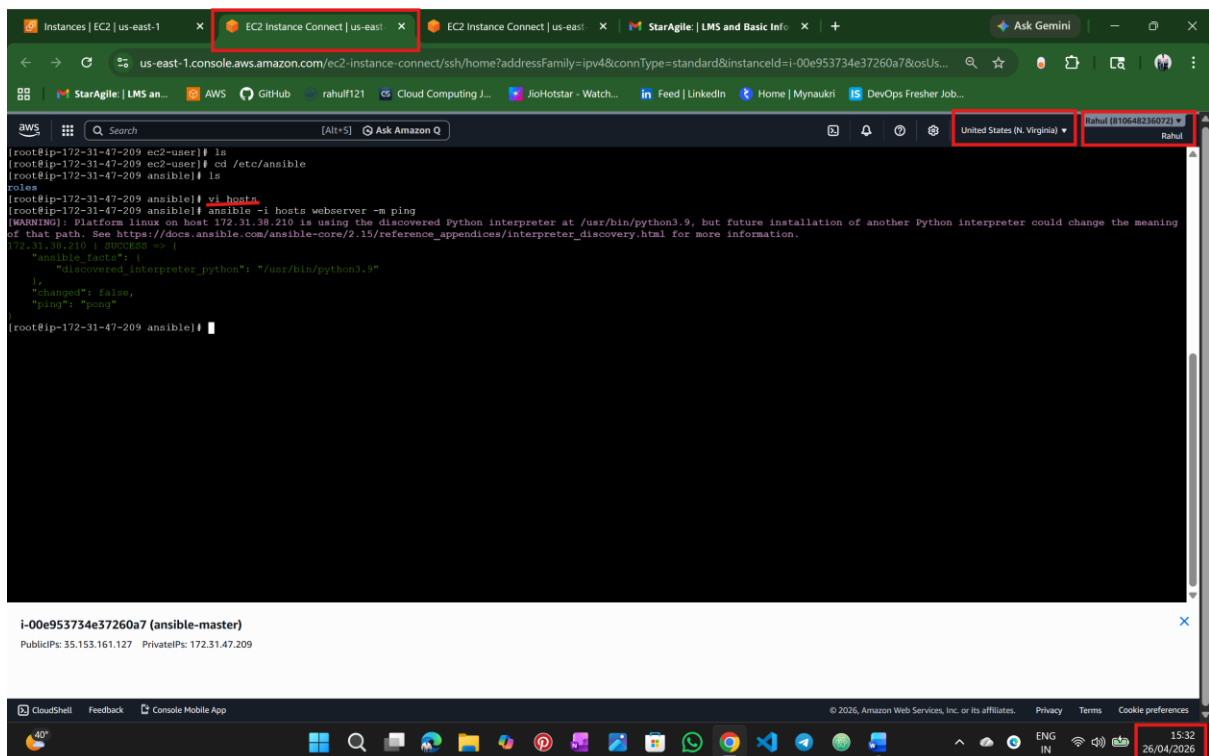
Step:4 this is worker node terminal paste the ssh-key in worker node



```
[root@ip-172-31-38-210 ec2-user]# cd .ssh
[root@ip-172-31-38-210 .ssh]# ls
authorized_keys
[root@ip-172-31-38-210 .ssh]# vim authorized_keys
[root@ip-172-31-38-210 .ssh]# chmod 700 ~/.ssh
[root@ip-172-31-38-210 .ssh]#
```

i-03db2b90fd717e5a (ansible-worker)
PublicPis: 3.91.88.159 PrivatePis: 172.31.38.210

Step:5 verify whether master node and worker node is connected are not? Yes! It successfully connected.



```
[root@ip-172-31-47-209 ec2-user]# ls
[root@ip-172-31-47-209 ec2-user]# cd /etc/ansible
[root@ip-172-31-47-209 ansible]# ls
roles
[root@ip-172-31-47-209 ansible]# vi hosts
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.38.210 ansible-hosts=1
  ansible_facts: {}
  "discovered_interpreter_python": "/usr/bin/python3.9"
}
"changed": false,
"ping": "pong"
[root@ip-172-31-47-209 ansible]#
```

i-00e953734e37260a7 (ansible-master)
PublicPis: 35.153.161.127 PrivatePis: 172.31.47.209

Step:6 Now, write the playbook yaml file to install java and maven

The screenshot shows the AWS CloudShell interface. The terminal displays the following Ansible playbook content:

```
--
- name: Setup Java Maven Build Server
  hosts: webserver
  become: yes

  tasks:
    - name: Install Java
      yum:
        name: java-17-amazon-corretto
        state: present
    - name: Install Maven
      yum:
        name: maven
        state: present
```

The interface includes a top navigation bar with tabs for 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'StarAgile: LMS and Basic Info'. The right sidebar shows the 'United States (N. Virginia)' region and the user 'Rahul (810648236072)'. The bottom status bar indicates the instance 'i-00e953734e37260a7 (ansible-master)' with public IP 35.153.161.127 and private IP 172.31.47.209. The system clock shows 15:37 on 26/04/2026.

Step:7 Run the playbook yaml file

The screenshot shows the AWS CloudShell interface with the Ansible playbook being executed. The terminal output is as follows:

```
[root@ip-172-31-47-209 ansible] vim playbook.yml
[root@ip-172-31-47-209 ansible] ansible-playbook -i hosts playbook.yml

PLAY [Setup Java Maven Build Server] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.38.210]

TASK [Install Java] *****
changed: [172.31.38.210]

TASK [Install Maven] *****
ok: [172.31.38.210]

PLAY RECAP *****
172.31.38.210      : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[root@ip-172-31-47-209 ansible] █
```

The interface is identical to the previous screenshot, showing the same navigation tabs, sidebar, and bottom status bar with the instance 'i-00e953734e37260a7 (ansible-master)' and system clock at 15:40 on 26/04/2026.

Step:8 Result, check and verify,

The screenshot shows a web browser window with the AWS console. The active tab is 'EC2 Instance Connect | us-east-1'. The URL is 'us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-03db2b90df717e5a&osUser=...'. The terminal window shows the following commands and output:

```
[root@ip-172-31-38-210 ec2-user]# cd .ssh
[root@ip-172-31-38-210 .ssh]# ls
authorized_keys
[root@ip-172-31-38-210 .ssh]# vim authorized_keys
[root@ip-172-31-38-210 .ssh]# chmod 700 ~/.ssh
[root@ip-172-31-38-210 .ssh]# chmod 600 ~/.ssh/authorized_keys
[root@ip-172-31-38-210 .ssh]#
[root@ip-172-31-38-210 .ssh]#
[root@ip-172-31-38-210 .ssh]# java -version
openjdk version "17.0.18" 2024-01-20 LTS
OpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.18.9.1 (build 17.0.18+9-LTS, mixed mode, sharing)
[root@ip-172-31-38-210 .ssh]# mvn -version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.18, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.166-197.305.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@ip-172-31-38-210 .ssh]#
```

Below the terminal, the instance details are shown: 'i-03db2b90df717e5a (ansible-worker)' with Public IP: 3.91.88.159 and Private IP: 172.31.38.210.

The bottom of the screenshot shows the Windows taskbar with the date and time '15:43 26/04/2026' highlighted in a red box.

Nalathathi Rahul